

WEEK 4: ORDER MANAGEMENT SYSTEM

PET SHOP MANAGEMENT SYSTEM

This document adapts the Week 4 Order Management System task to a Pet Shop Management System. It follows the structure of the provided task: Orders and Order_Details tables, relationships, constraints, order management operations, and reports.

1. Design Orders and Order_Details Tables

Orders Entity – Stores overall order information.

Attribute	Description
Order_ID	Unique order identifier
Customer_ID	Customer reference
Order_Date	Date of order
Total_Amount	Total bill amount
Order_Status	Pending/Shipped/Delivered

SQL Code:

```
SQL> CREATE TABLE OrdersEntity (  
2     Order_ID VARCHAR2(7),  
3     Customer_ID VARCHAR2(7),  
4     Order_Date DATE,  
5     Total_Amount NUMBER(9,2),  
6     Order_Status VARCHAR2(15),  
7     PRIMARY KEY (Order_ID),  
8     FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)  
9 );
```

Table created.

Order_Details Entity – Stores pets/shop products included in an order.

Attribute	Description
Order_Detail_ID	Unique detail identifier
Order_ID	Order reference
Product_ID	Pet shop product reference
Quantity	Number of products purchased
Price	Product price

```

SQL> CREATE TABLE OrderDetail (
  2     Order_Detail_ID VARCHAR2(7),
  3     Order_ID VARCHAR2(7),
  4     Product_ID VARCHAR2(7),
  5     Quantity NUMBER,
  6     Price NUMBER(9,2),
  7     PRIMARY KEY (Order_Detail_ID),
  8     FOREIGN KEY (Order_ID) REFERENCES OrdersEntity(Order_ID),
  9     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
10 );

Table created.

```

2. Establish Relationships

Customer and Orders

- One customer can place multiple orders.
- Each order belongs to one customer.

Relationship: Customer (1) ----- (Many) Orders

Orders and Products

- One order can contain multiple products.
- One product can appear in multiple orders.

Relationship: Orders (Many) ----- (Many) Product

This many-to-many relationship is implemented using the OrderDetail table.

3. Apply Constraints

```

SQL> ALTER TABLE OrdersEntity
  2  MODIFY Customer_ID NOT NULL;

Table altered.

```

```

SQL> ALTER TABLE OrdersEntity
  2  MODIFY Order_Date NOT NULL;

Table altered.

```

```

SQL> ALTER TABLE OrdersEntity
  2  ADD CONSTRAINT chk_order_status
  3  CHECK (Order_Status IN ('Pending', 'Shipped', 'Delivered'));

Table altered.

```

```
SQL> ALTER TABLE OrderDetail
  2  MODIFY Order_ID NOT NULL;

Table altered.
```

```
SQL> ALTER TABLE OrderDetail
  2  MODIFY Product_ID NOT NULL;

Table altered.
```

```
SQL> ALTER TABLE OrderDetail
  2  ADD CONSTRAINT chk_quantity
  3  CHECK (Quantity > 0);

Table altered.
```

```
ALTER TABLE OrderDetail
ADD CONSTRAINT chk_price
CHECK (Price >= 0);
```

4. Insert 10 Values

Orders Table:

```
SQL> INSERT INTO OrdersEntity
  2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3  VALUES
  4  ('01001', 'S101', DATE '2026-01-05', 1250.50, 'Delivered');

1 row created.
```

```
SQL>
SQL> INSERT INTO OrdersEntity
  2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3  VALUES
  4  ('01002', 'S102', DATE '2026-01-08', 850.00, 'Shipped');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3  VALUES
  4  ('01004', 'S104', DATE '2026-01-15', 450.25, 'Delivered');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01006', 'S106', DATE '2026-02-02', 999.99, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01007', 'S107', DATE '2026-02-10', 3200.50, 'Delivered');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01009', 'S109', DATE '2026-02-25', 1450.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01010', 'S110', DATE '2026-03-01', 2750.80, 'Shipped');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01003', 'S103', DATE '2026-01-12', 2300.75, 'Pending');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01005', 'S105', DATE '2026-01-20', 1750.00, 'Pending');

1 row created.
```

```
SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('01008', 'S108', DATE '2026-02-18', 675.25, 'Pending');

1 row created.
```

ORDER_I	CUSTOME	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
01001	S101	05-JAN-26	1250.5	Delivered
01002	S102	08-JAN-26	850	Shipped
01004	S104	15-JAN-26	450.25	Delivered
01006	S106	02-FEB-26	999.99	Shipped
01007	S107	10-FEB-26	3200.5	Delivered
01009	S109	25-FEB-26	1450	Delivered
01010	S110	01-MAR-26	2750.8	Shipped
01003	S103	12-JAN-26	2300.75	Pending
01005	S105	20-JAN-26	1750	Pending
01008	S108	18-FEB-26	675.25	Pending

10 rows selected.

OrderDetail Table:

```
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1001', 'O1001', 'P101', 2, 550.00);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1002', 'O1002', 'P102', 1, 450.00);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1003', 'O1003', 'P103', 3, 180.40);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1004', 'O1004', 'P101', 1, 550.00);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1005', 'O1005', 'P102', 2, 450.00);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1006', 'O1006', 'P103', 4, 180.40);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1007', 'O1007', 'P101', 3, 550.00);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1008', 'O1008', 'P102', 2, 450.00);

1 row created.
```

```

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1009', 'O1009', 'P103', 5, 180.40);

1 row created.

SQL>
SQL> INSERT INTO OrderDetail
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3 VALUES
  4 ('OD1010', 'O1010', 'P101', 2, 550.00);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.

```

```

SQL> select * from orderdetail;

```

ORDER_D	ORDER_I	PRODUCT	QUANTITY	PRICE
OD1001	O1001	P101	2	550
OD1002	O1002	P102	1	450
OD1003	O1003	P103	3	180.4
OD1004	O1004	P101	1	550
OD1005	O1005	P102	2	450
OD1006	O1006	P103	4	180.4
OD1007	O1007	P101	3	550
OD1008	O1008	P102	2	450
OD1009	O1009	P103	5	180.4
OD1010	O1010	P101	2	550

```

10 rows selected.

```

5. Manage Customer Product Orders

Create a new order

```

SQL> INSERT INTO OrdersEntity
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)
  3 VALUES
  4 ('O1011', 'S101', SYSDATE, 1500.00, 'Pending');

1 row created.

```

Add product to an order

```
SQL> INSERT INTO OrderDetail
  2  (Order_Detail_ID, Order_ID, Product_ID, Quantity, Price)
  3  VALUES
  4  ('OD1011', '01011', 'P101', 3, 500.00);

1 row created.
```

View complete order details

```

SQL> SELECT o.Order_ID,
2         o.Customer_ID,
3         o.Order_Date,
4         od.Product_ID,
5         od.Quantity,
6         od.Price,
7         o.Total_Amount,
8         o.Order_Status
9 FROM OrdersEntity o
10 JOIN OrderDetail od
11 ON o.Order_ID = od.Order_ID;

```

ORDER_I	CUSTOME	ORDER_DAT	PRODUCT	QUANTITY	PRICE	TOTAL_AMOUNT
ORDER_STATUS						
01001	S101	05-JAN-26	P101	2	550	1250.5
Delivered						

01002	S102	08-JAN-26	P102	1	450	850
Shipped						

01003	S103	12-JAN-26	P103	3	180.4	2300.75
Pending						

ORDER_I	CUSTOME	ORDER_DAT	PRODUCT	QUANTITY	PRICE	TOTAL_AMOUNT
ORDER_STATUS						
01004	S104	15-JAN-26	P101	1	550	450.25
Delivered						

01005	S105	20-JAN-26	P102	2	450	1750
Pending						

01006	S106	02-FEB-26	P103	4	180.4	999.99
Shipped						

ORDER_I	CUSTOME	ORDER_DAT	PRODUCT	QUANTITY	PRICE	TOTAL_AMOUNT
ORDER_STATUS						
01007	S107	10-FEB-26	P101	3	550	3200.5
Delivered						

01008	S108	18-FEB-26	P102	2	450	675.25
Pending						

01009	S109	25-FEB-26	P103	5	180.4	1450
Delivered						

ORDER_I	CUSTOME	ORDER_DAT	PRODUCT	QUANTITY	PRICE	TOTAL_AMOUNT
-----	-----	-----	-----	-----	-----	-----
ORDER_STATUS						

01010	S110	01-MAR-26	P101	2	550	2750.8
Shipped						
01011	S101	05-OCT-26	P101	3	500	1500
Pending						
11 rows selected.						

6. Update Operation

```
SQL> UPDATE OrdersEntity
  2 SET Order_Status = 'Delivered'
  3 WHERE Order_ID = '01011';

1 row updated.

SQL>
SQL> UPDATE OrderDetail
  2 SET Quantity = 5
  3 WHERE Order_Detail_ID = 'OD1011';

1 row updated.
```

7. Delete Operation

```
SQL> DELETE FROM OrderDetail
  2 WHERE Order_ID = '01011';

1 row deleted.

SQL> DELETE FROM OrdersEntity
  2 WHERE Order_ID = '01011';

1 row deleted.
```

8. Customer Order History Report

```

SQL> SELECT c.Customer_ID,
2      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
3      o.Order_ID,
4      o.Order_Date,
5      o.Total_Amount,
6      o.Order_Status
7 FROM Customer c
8 JOIN OrdersEntity o
9 ON c.Customer_ID = o.Customer_ID
10 ORDER BY o.Order_Date;

```

CUSTOMER_ID		ORDER_I	ORDER_DAT
S101	Akshaya Nivasini 1250.5 Delivered	01001	05-JAN-26
S102	Prithish Raj 850 Shipped	01002	08-JAN-26
S103	kshaunish Rajesh 2300.75 Pending	01003	12-JAN-26
S104	kaniska Ram	01004	15-JAN-26
S105	Rukmani Kuppusamy 1750 Pending	01005	20-JAN-26
S106			

```

CUSTOMER_ID
-----
CUSTOMER_NAME                                ORDER_I ORDER_DAT
-----
TOTAL_AMOUNT ORDER_STATUS
-----
Vijay Deepika                                01006   02-FEB-26
    999.99 Shipped

S107
Aura Nivas                                    01007   10-FEB-26
    3200.5 Delivered

CUSTOMER_ID
-----
CUSTOMER_NAME                                ORDER_I ORDER_DAT
-----
TOTAL_AMOUNT ORDER_STATUS
-----
S108
Shreya Raman                                01008   18-FEB-26
    675.25 Pending

S109
Geetha Govind                                01009   25-FEB-26
    1450 Delivered

CUSTOMER_ID
-----
CUSTOMER_NAME                                ORDER_I ORDER_DAT
-----
TOTAL_AMOUNT ORDER_STATUS
-----
S110
Ramya Krishnan                              01010   01-MAR-26
    2750.8 Shipped

10 rows selected.

```

9. Product-wise Order Report

```

SQL> SELECT p.Product_ID,
2         p.Product_Name,
3         COUNT(od.Order_ID) AS Times_Ordered,
4         SUM(od.Quantity) AS Total_Quantity_Sold
5 FROM Product p
6 JOIN OrderDetail od
7 ON p.Product_ID = od.Product_ID
8 GROUP BY p.Product_ID, p.Product_Name
9 ORDER BY Total_Quantity_Sold DESC;

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
TIMES_ORDERED TOTAL_QUANTITY_SOLD
-----
P103
bird seed mix
          3                12

```

```

P101
pedigree dog food
          4                8

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
TIMES_ORDERED TOTAL_QUANTITY_SOLD
-----
P102
cat dry food
          3                5

```

10. Customer Purchase Analysis

CUSTOMER_ID		

CUSTOMER_NAME		TOTAL_ORDERS

TOTAL_PURCHASE_AMOUNT		

S107		
Aura Nivas		1
	3200.5	
S110		
Ramya Krishnan		1
	2750.8	
CUSTOMER_ID		

CUSTOMER_NAME		TOTAL_ORDERS

TOTAL_PURCHASE_AMOUNT		

S103		
kshaunish Rajesh		1
	2300.75	
S105		
Rukmani Kuppusamy		1
CUSTOMER_ID		

CUSTOMER_NAME		TOTAL_ORDERS

TOTAL_PURCHASE_AMOUNT		

	1750	
S109		
Geetha Govind		1
	1450	
S101		
CUSTOMER_ID		

CUSTOMER_NAME		TOTAL_ORDERS

TOTAL_PURCHASE_AMOUNT		

Akshaya Nivasini		1
	1250.5	
S106		
Vijay Deepika		1
	999.99	

[illegible]